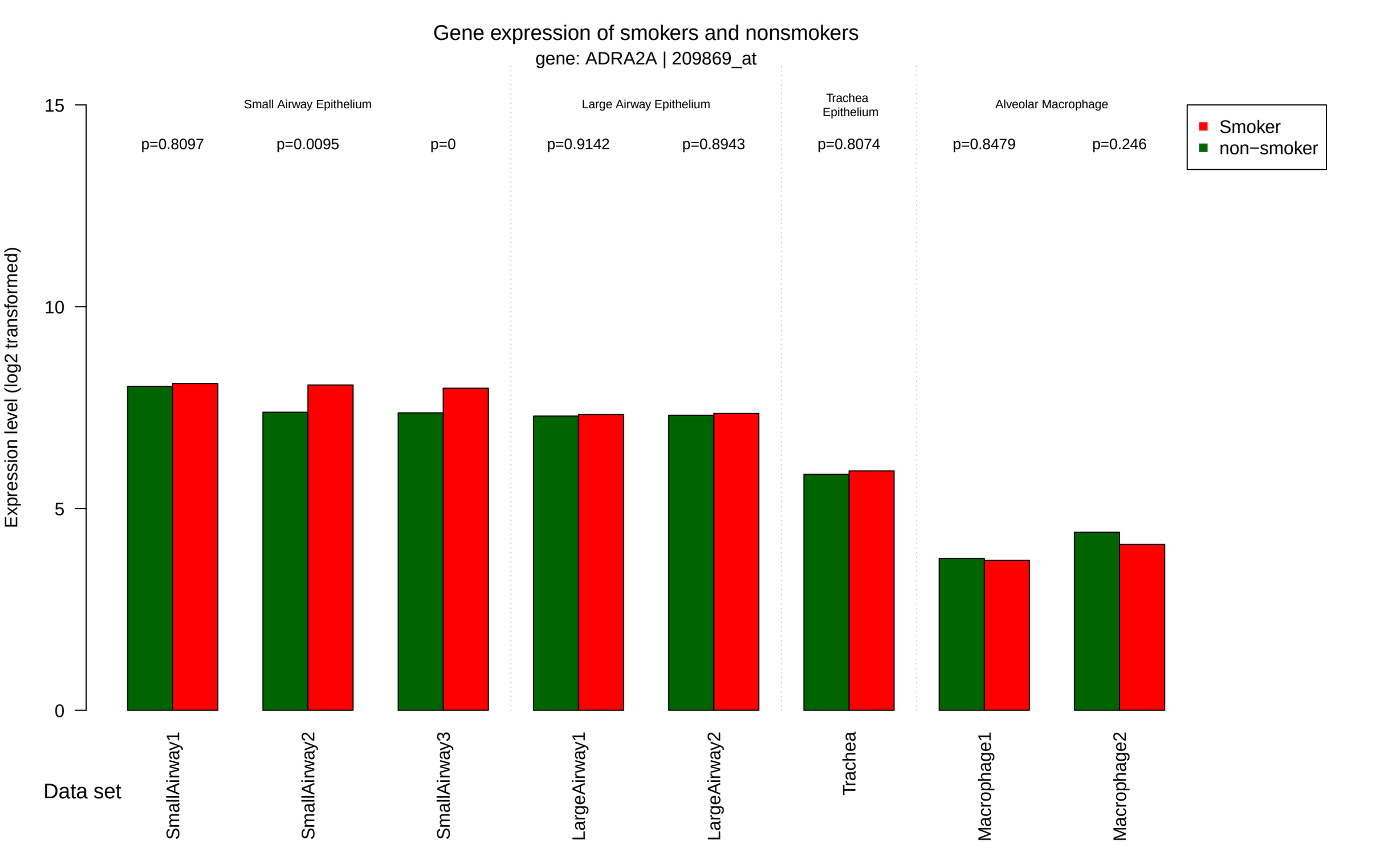
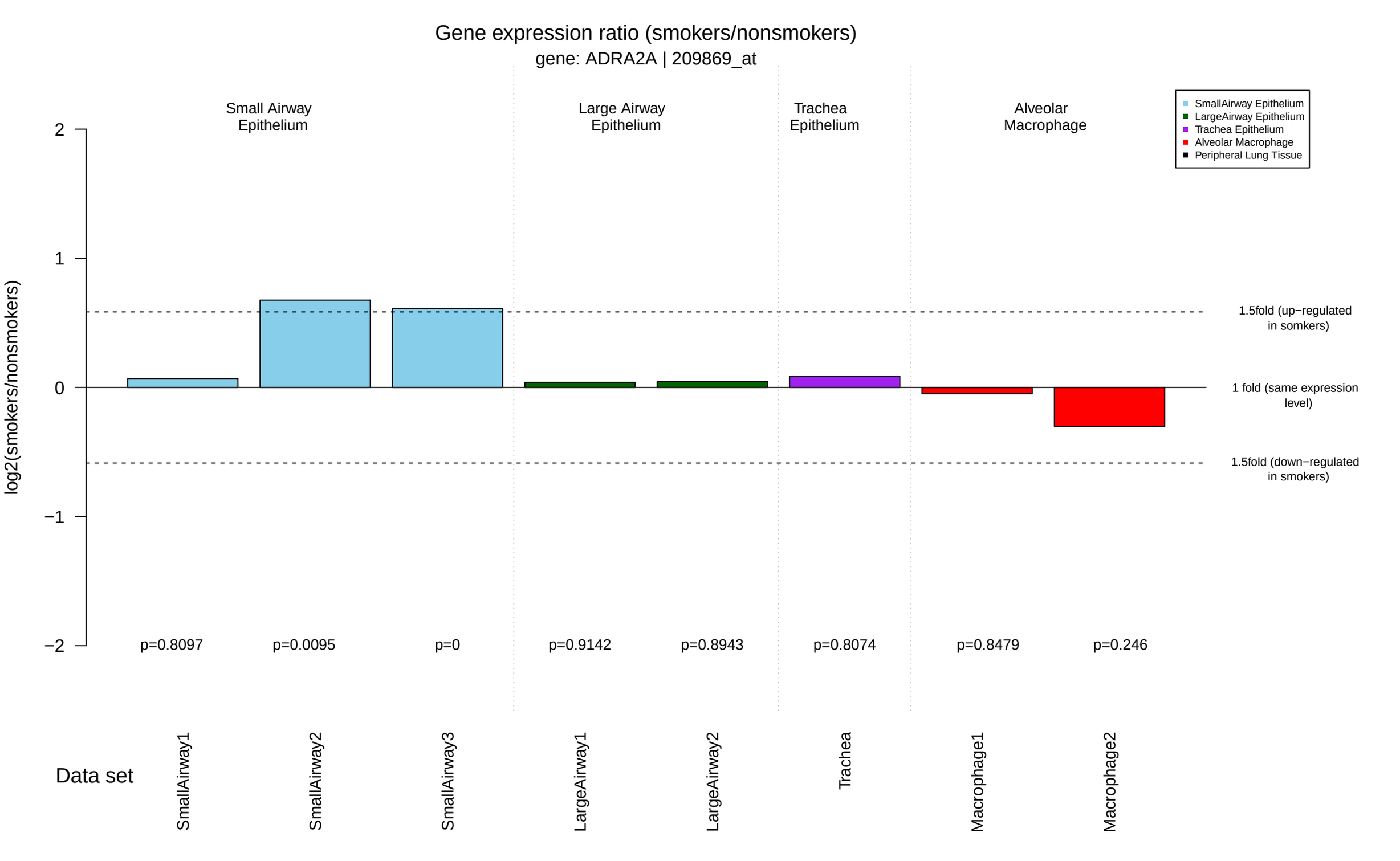
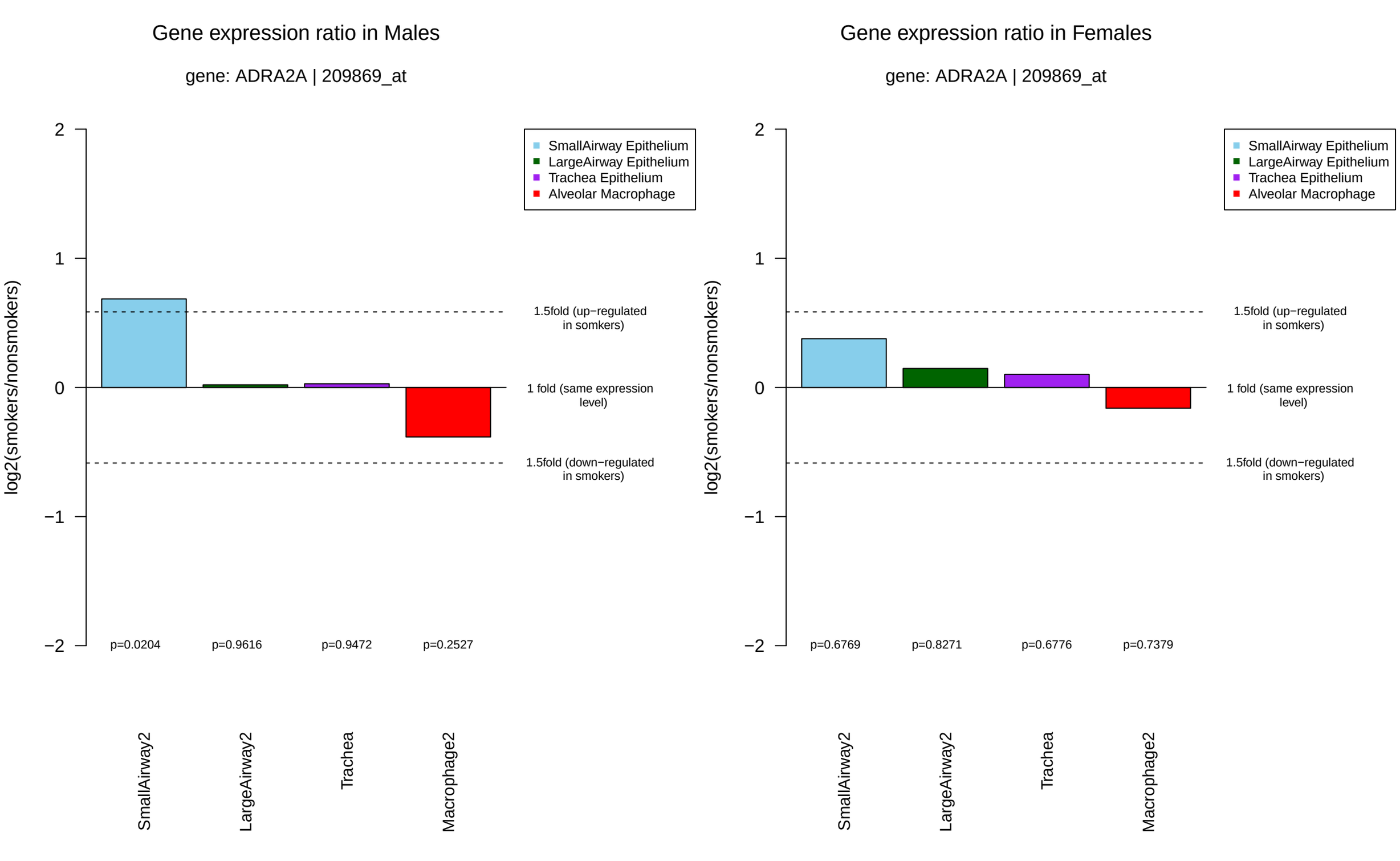
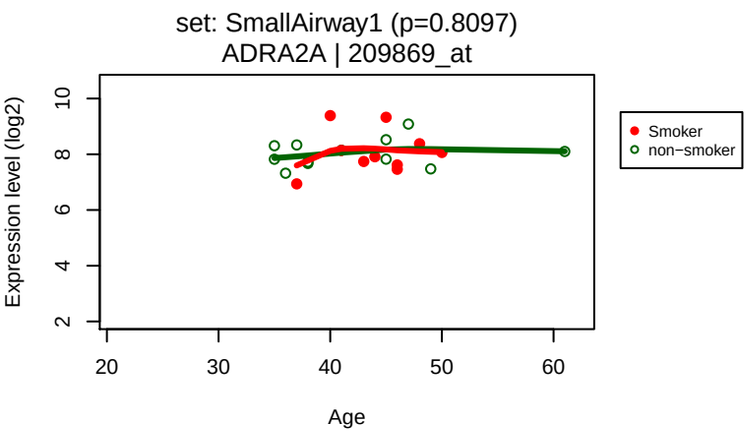
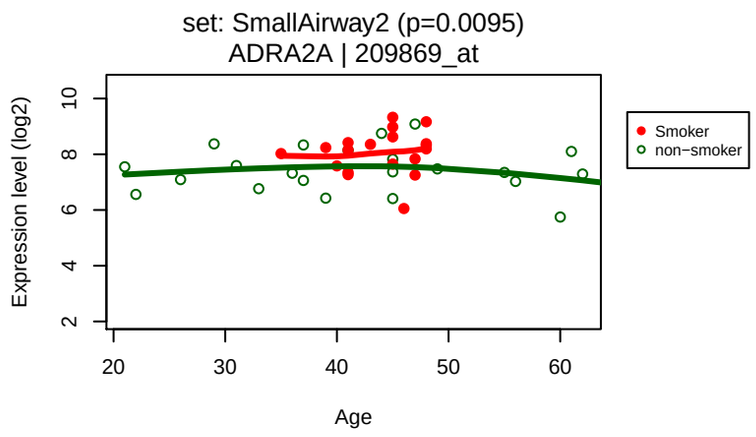
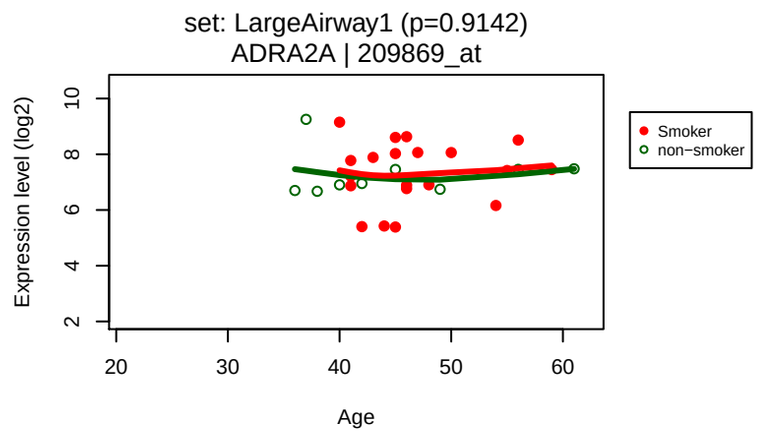

Gene expression of smokers and nonsmokers
gene: ADRA2A | 209869_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8097
p=0.0095
p=0
p=0.9142
p=0.8943
p=0.8074
p=0.8479
p=0.246
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

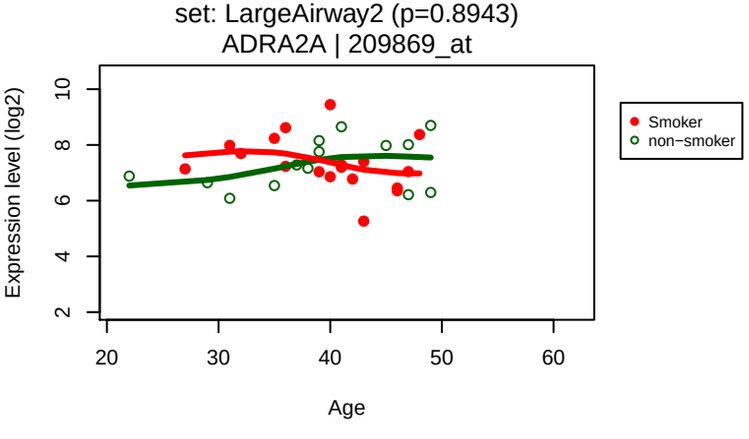

Gene expression ratio (smokers/nonsmokers)
gene: ADRA2A | 209869_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8097
p=0.0095
p=0
p=0.9142
p=0.8943
p=0.8074
p=0.8479
p=0.246
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

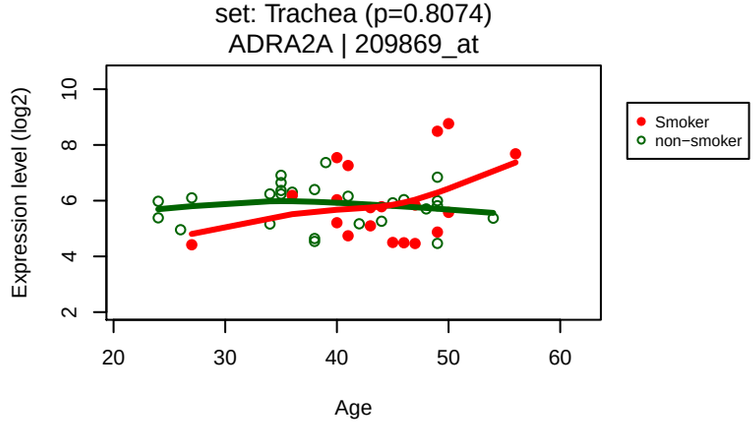

Gene expression ratio in Males
gene: ADRA2A | 209869_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0204
p=0.9616
p=0.9472
p=0.2527
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADRA2A | 209869_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6769
p=0.8271
p=0.6776
p=0.7379
SmallAirway2
LargeAirway2
Trachea
Macrophage2

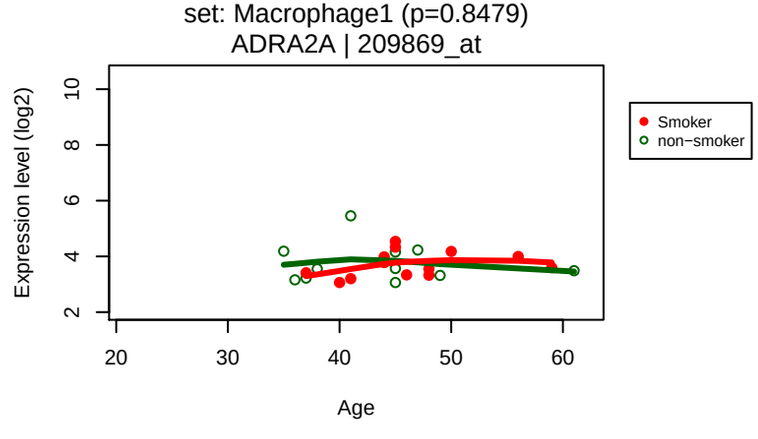

set: SmallAirway1 (p=0.8097)
ADRA2A | 209869_at
Expression level (log2)
Age
Smoker
non−smoker

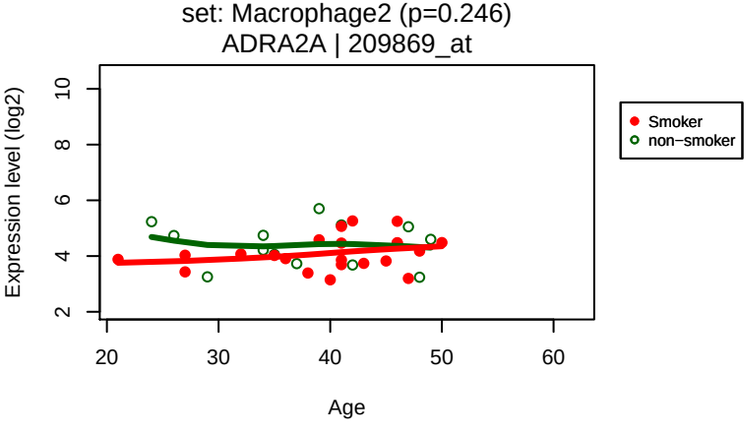

set: SmallAirway2 (p=0.0095)
ADRA2A | 209869_at
Expression level (log2)
Age
Smoker
non−smoker

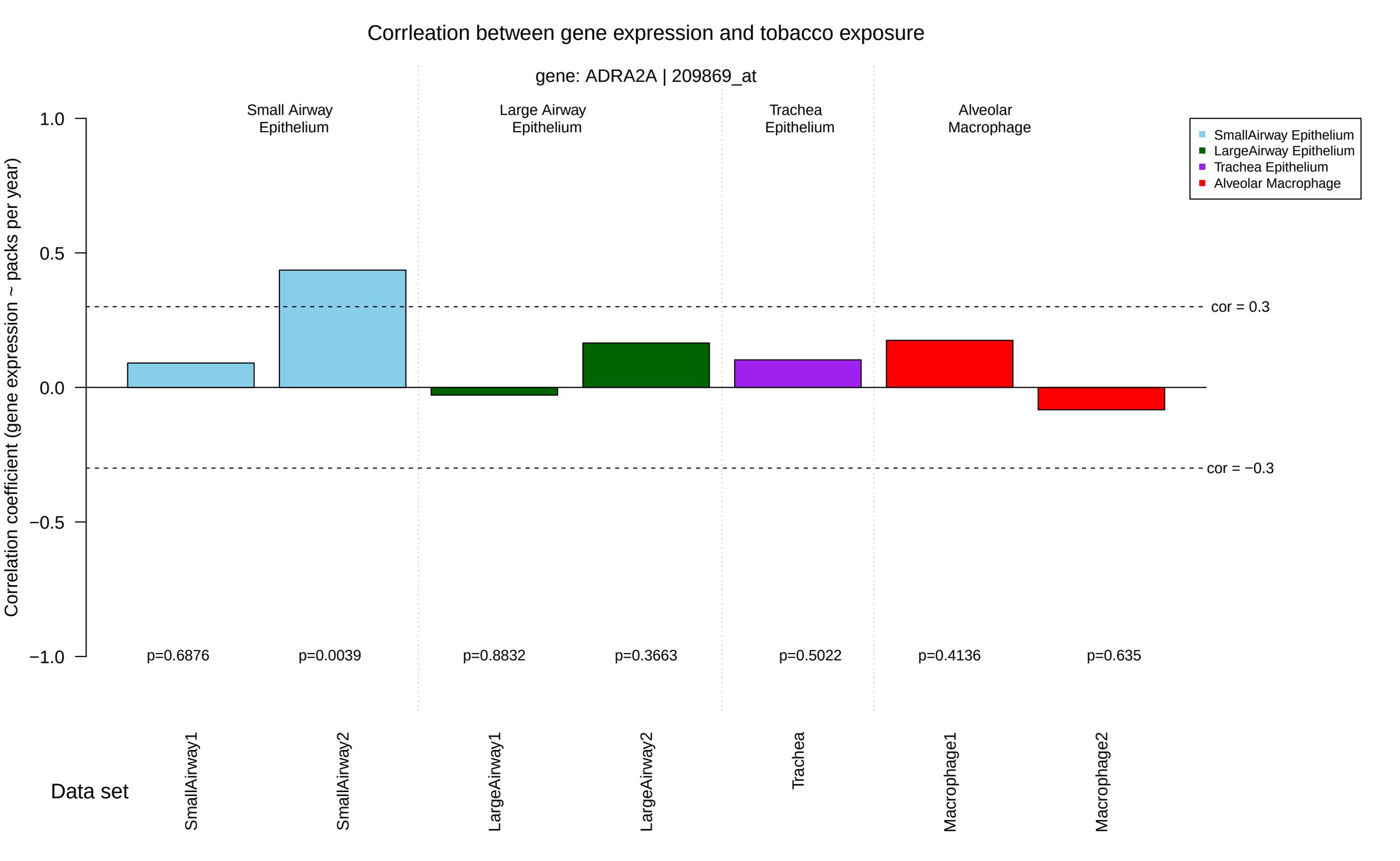

set: LargeAirway1 (p=0.9142)
ADRA2A | 209869_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8943)
ADRA2A | 209869_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8074)
ADRA2A | 209869_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8479)
ADRA2A | 209869_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.246)
ADRA2A | 209869_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADRA2A | 209869_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6876
p=0.0039
p=0.8832
p=0.3663
p=0.5022
p=0.4136
p=0.635
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2